Entanglement-Sensitive Observables for a Two-Qubit Quantum Sensor

Spin and Spatial Entanglement, Collective Observables, and Links to High-Energy Physics

Motivation

Consider two qubits used as a prototype quantum sensor. The qubits may be entangled in

- spin,
- spatial mode,
- or both.

A natural question is:

> *Which observables best reveal the role of entanglement in the sensing protocol?*

A second question is whether one can go beyond local measurements and use observables acting on the whole two-qubit system.

Basic Setup

Let the two qubits be labeled A and B. For spin-1/2 degrees of freedom, define Pauli operators

$$\sigma_x^{(A)}, \sigma_y^{(A)}, \sigma_z^{(A)}, \qquad \sigma_x^{(B)}, \sigma_y^{(B)}, \sigma_z^{(B)}.$$

We may also define collective operators

$$J_\alpha = \frac{1}{2}\left(\sigma_\alpha^{(A)} + \sigma_\alpha^{(B)}\right), \qquad \alpha = x, y, z.$$

If one qubit couples to a magnetic field B, a simple interaction Hamiltonian is

$$H_{\mathrm{int}} = \frac{\gamma B}{2}\sigma_z^{(A)}.$$

A more symmetric model is

$$H_{\mathrm{int}} = \frac{\gamma_A B_A}{2}\sigma_z^{(A)} + \frac{\gamma_B B_B}{2}\sigma_z^{(B)}.$$

Canonical Entangled Sensor States

Typical two-qubit entangled states include the Bell states

$$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \tag{1}$$

$$|\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle). \tag{2}$$

For sensing, these states are important because they respond differently to common-mode and differential fields.
Examples:

- $|\Phi^{\pm}\rangle$ are sensitive to collective phase accumulation,
- $|\Psi^{\pm}\rangle$ can be sensitive to phase gradients and differential couplings.

Local Observables

The most obvious observables are local spin components:

$$\left\langle \sigma_\alpha^{(A)} \right\rangle, \qquad \left\langle \sigma_\alpha^{(B)} \right\rangle.$$

If only qubit A is exposed to the signal field, then

$$\left\langle \sigma_z^{(A)} \right\rangle, \quad \left\langle \sigma_x^{(A)} \right\rangle, \quad \left\langle \sigma_y^{(A)} \right\rangle$$

can reveal local phase accumulation.
However, these local observables do not always capture the metrological value of entanglement. In many cases, entanglement is invisible in all one-body marginals but appears clearly in correlations.

Two-Body Correlation Observables

The simplest genuinely nonlocal observables are correlators

$$C_{\alpha\beta} = \left\langle \sigma_\alpha^{(A)} \sigma_\beta^{(B)} \right\rangle, \qquad \alpha, \beta \in \{x, y, z\}.$$

Important examples:

$$\left\langle \sigma_z^{(A)} \sigma_z^{(B)} \right\rangle, \tag{3}$$

$$\left\langle \sigma_x^{(A)} \sigma_x^{(B)} \right\rangle, \tag{4}$$

$$\left\langle \sigma_y^{(A)} \sigma_y^{(B)} \right\rangle. \tag{5}$$

These observables:

- distinguish Bell states,
- quantify alignment/anti-alignment,
- detect how phase information is encoded jointly in the two qubits.

They are often the most direct observables for testing whether entanglement improves sensing.

Connected Correlators

A better diagnostic than the raw correlator is often the connected correlation

$$\langle \sigma_\alpha^{(A)} \sigma_\beta^{(B)} \rangle_c = \left\langle \sigma_\alpha^{(A)} \sigma_\beta^{(B)} \right\rangle - \left\langle \sigma_\alpha^{(A)} \right\rangle \left\langle \sigma_\beta^{(B)} \right\rangle .$$

Why useful?

- It removes trivial product-state contributions.
- It isolates genuine correlation effects.
- It is often nonzero precisely when entanglement is operationally relevant.

In sensing, a strong field dependence of connected correlators can signal that the parameter is encoded nonlocally.

Collective Spin Observables

Observables acting on the whole system include

$$J_x, \qquad J_y, \qquad J_z$$

and quadratic forms

$$J_x^2, \qquad J_y^2, \qquad J_z^2.$$

For two qubits,

$$J_\alpha^2 = \frac{1}{2}\left(I + \sigma_\alpha^{(A)}\sigma_\alpha^{(B)}\right).$$

Thus collective spin fluctuations directly probe two-body correlations.

Important quantities:

$$\langle J_\alpha \rangle, \tag{6}$$

$$(\Delta J_\alpha)^2 = \left\langle J_\alpha^2 \right\rangle - \langle J_\alpha \rangle^2. \tag{7}$$

These are natural observables for testing whether entanglement sharpens or broadens phase sensitivity.

Parity Observables

A very important global observable is parity. For example, along the z axis:

$$\Pi_z = \sigma_z^{(A)} \sigma_z^{(B)}.$$

More generally, after a collective rotation one can measure

$$\Pi(\phi) = U^\dagger(\phi)\, \sigma_z^{(A)} \sigma_z^{(B)}\, U(\phi).$$

Parity is central in quantum metrology because

- it is highly sensitive to phase accumulation,
- it can saturate optimal sensitivity in some entangled interferometers,
- it is directly measurable in many qubit platforms.

For Bell-like states, parity oscillations are often a clean signature of entanglement-enhanced sensing.

Bell Operators and CHSH-Type Observables

Another nonlocal observable class is the Bell operator, e.g.

$$\mathcal{B} = \mathbf{a} \cdot \boldsymbol{\sigma}^{(A)} \otimes (\mathbf{b} + \mathbf{b}') \cdot \boldsymbol{\sigma}^{(B)} + \mathbf{a}' \cdot \boldsymbol{\sigma}^{(A)} \otimes (\mathbf{b} - \mathbf{b}') \cdot \boldsymbol{\sigma}^{(B)}.$$

Its expectation value probes Bell nonlocality.
Why relevant for sensing?

- Bell-operator expectation values are built from precisely the correlators that encode nonclassical joint response.
- If the signal changes $\langle \mathcal{B} \rangle$, then the sensing process is affecting nonlocal quantum correlations, not merely local polarization.

This is especially useful when studying robustness of sensing advantage against decoherence.

Quantum Fisher Information as the Central Quantity

For parameter estimation, the most important quantity is often the quantum Fisher information (QFI),

$$F_Q[\rho, G],$$

where G is the generator of the parameter encoding.
For pure states,

$$F_Q = 4(\Delta G)^2.$$

Thus the observable content is encoded in the variance of the generator.
Examples:

- If the field couples only to qubit A, then $G = \sigma_z^{(A)}/2$.
- If the field couples collectively, then $G = J_z$.
- If one probes a gradient field, then $G = (\sigma_z^{(A)} - \sigma_z^{(B)})/2$.

Entanglement helps when it enhances the variance of the relevant global generator.

Observables for Uniform and Gradient Fields

Two especially useful whole-system generators are

$$G_{\mathrm{com}} = \frac{1}{2}\left(\sigma_z^{(A)} + \sigma_z^{(B)}\right) = J_z$$

for common-mode fields, and

$$G_{\mathrm{diff}} = \frac{1}{2}\left(\sigma_z^{(A)} - \sigma_z^{(B)}\right)$$

for gradient or differential fields.
Then one studies

$$(\Delta G_{\mathrm{com}})^2, \qquad (\Delta G_{\mathrm{diff}})^2.$$

Different entangled states optimize different sensing tasks:

- $|\Phi^{\pm}\rangle$ can be highly sensitive to common phase shifts,
- $|\Psi^{\pm}\rangle$ can be highly sensitive to differential phase shifts.

If the Qubits Are Also Spatially Entangled

Suppose each qubit also has a spatial degree of freedom, for example two paths or two locations. Then we may define mode operators or effective spatial Pauli operators

$$\tau_x, \tau_y, \tau_z$$

acting on the spatial subspace.

Then one can study hybrid observables such as

$$\sigma_z^{(A)} \otimes \tau_z^{(A)}, \qquad \sigma_z^{(A)} \sigma_z^{(B)}, \qquad \tau_x^{(A)} \tau_x^{(B)},$$

or mixed spin-space correlators

$$\sigma_x^{(A)} \tau_x^{(A)} \, \sigma_x^{(B)} \tau_x^{(B)}.$$

These observables probe whether the sensing advantage arises from

- spin entanglement,
- spatial entanglement,
- or spin-space hybrid entanglement.

Total-Spin Observables

For two spin-$1/2$ qubits, another very natural observable is

$$\mathbf{J}^2 = J_x^2 + J_y^2 + J_z^2.$$

This distinguishes singlet and triplet sectors:

$$\mathbf{J}^2 \left|\text{triplet}\right\rangle = 2 \left|\text{triplet}\right\rangle, \tag{8}$$

$$\mathbf{J}^2 \left|\text{singlet}\right\rangle = 0. \tag{9}$$

This is useful because:

- the singlet state is rotationally invariant,
- triplet states respond differently to external fields,
- total-spin measurements reveal how symmetry constrains sensing response.

This observable acts on the whole two-qubit system and is often more informative than local spin projections alone.

Entanglement Witnesses as Observables

One may also construct entanglement witnesses W such that

$$\langle W \rangle < 0$$

signals entanglement.
For example, witnesses built from correlators

$$W \sim I - \sigma_x^{(A)}\sigma_x^{(B)} - \sigma_y^{(A)}\sigma_y^{(B)} - \sigma_z^{(A)}\sigma_z^{(B)}$$

can be directly measurable.
These are not always optimal sensing observables, but they are useful for

- verifying that the enhanced sensitivity truly comes from entanglement,
- separating entanglement-enhanced performance from classical correlation effects.

Which Observables Are Best for Testing Entanglement in Sensing?

A practical hierarchy is:

1. Local magnetization:
$$\left\langle \sigma_z^{(A)} \right\rangle, \quad \left\langle \sigma_z^{(B)} \right\rangle$$
2. Two-body correlators:
$$\left\langle \sigma_\alpha^{(A)} \sigma_\beta^{(B)} \right\rangle$$
3. Collective observables:
$$\langle J_\alpha \rangle, \quad (\Delta J_\alpha)^2, \quad \langle \mathbf{J}^2 \rangle$$
4. Parity:
$$\left\langle \sigma_z^{(A)} \sigma_z^{(B)} \right\rangle$$
5. Generator variances / QFI:
$$(\Delta G)^2, \quad F_Q$$

If the goal is specifically to quantify metrological gain, then generator variance and QFI are the most directly relevant

Relevance for High-Energy Physics

Yes, several of these observables are relevant in high-energy physics.

Examples include:

- spin-correlation observables in relativistic two-particle production,
- Bell-type correlators for particle-antiparticle systems,
- flavor and spin oscillation observables,
- collective symmetry generators in quantum field theory,
- detector-response observables in relativistic quantum information.

The same mathematical structures appear:

$$\left\langle \sigma_\alpha^{(A)} \sigma_\beta^{(B)} \right\rangle, \qquad \langle \mathcal{B} \rangle, \qquad (\Delta G)^2.$$

Thus the language of quantum sensing overlaps strongly with the language of correlation measurements in high-energy and relativistic physics.

Examples Relevant to High-Energy Physics

Some concrete directions where these observables matter:

- **Top-quark or lepton spin correlations:** measured through correlation tensors analogous to

$$\left\langle \sigma_i^{(A)} \sigma_j^{(B)} \right\rangle .$$

- **Neutral meson or particle-antiparticle entanglement:** Bell-like observables and parity-type observables test quantum coherence.
- **Neutrino oscillation and flavor interferometry:** collective generators and phase observables resemble sensing generators.
- **Unruh/Hawking detector models:** two-detector correlations and entanglement harvesting use very similar joint observables.

So the same observables that diagnose entanglement-enhanced sensing can also be reinterpreted as probes of relativistic quantum correlations.

A Useful Minimal Measurement Program

For a two-qubit entangled quantum sensor, a minimal but powerful set of measurements is:

$$\left\langle \sigma_z^{(A)} \right\rangle, \qquad \left\langle \sigma_z^{(B)} \right\rangle, \tag{10}$$

$$\left\langle \sigma_x^{(A)} \sigma_x^{(B)} \right\rangle, \qquad \left\langle \sigma_y^{(A)} \sigma_y^{(B)} \right\rangle, \qquad \left\langle \sigma_z^{(A)} \sigma_z^{(B)} \right\rangle, \tag{11}$$

$$\langle J_z \rangle, \qquad (\Delta J_z)^2, \qquad (\Delta G_{\mathrm{diff}})^2. \tag{12}$$

If spatial entanglement is also present, add

$$\left\langle \tau_x^{(A)} \tau_x^{(B)} \right\rangle, \qquad \left\langle \tau_z^{(A)} \tau_z^{(B)} \right\rangle,$$

and mixed spin-space correlators.

This set already gives a strong picture of whether the advantage comes from local response, joint response, or hybrid spin-space entanglement.

Summary

For two spin and spatially entangled qubits used as a quantum sensor:

- Local observables probe direct coupling to the field.
- Two-body correlators reveal genuinely entangled response.
- Collective observables such as J_α, J_α^2, and $\mathbf{J}^2$ act on the whole system and are highly informative.
- Parity and Bell-type observables are powerful nonlocal diagnostics.
- The most fundamental metrological quantity is the QFI, determined by variances of the encoding generator.

Many of these observables are also directly relevant in high-energy physics, especially in studies of spin correlations, Bell tests, relativistic detector models, and quantum-field-theoretic interferometry.

Take-Home Message

If entanglement is central to the sensing protocol, then measuring only

$$\left\langle \sigma_z^{(A)} \right\rangle$$

is usually not enough.
The most informative whole-system observables are often

$$\left\langle \sigma_\alpha^{(A)} \sigma_\beta^{(B)} \right\rangle, \qquad \langle J_\alpha \rangle, \qquad \left\langle \mathbf{J}^2 \right\rangle, \qquad \langle \Pi \rangle, \qquad F_Q.$$

These capture how the parameter is encoded collectively and whether entanglement genuinely improves sensitivity.